

Zeeshan Manzoor Bhat

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ORCID: 0000-0003-2838-1151

Research Interests

Large-scale experimental testing; Nonlinear modelling of structures; Seismic hazard, risk and vulnerability assessment; Performance-based earthquake engineering; Resilience-based seismic design.

Education

Ph.D. in Structural and Earthquake Engineering

July 2018 – March 2025

Indian Institute of Technology Roorkee, India

Thesis: *Seismic Performance of Masonry Infilled RC Frame Buildings Designed for Modern Codes*

Advisors: Prof. Yogendra Singh, Prof. Pankaj Agarwal

M.Tech. in Structural Engineering

July 2016 – July 2018

National Institute of Technology Srinagar, India

Thesis: *An Experimental Investigation into the Behavior of Steel–Timber Composite Beams*

Advisor: Prof. Javed Ahmad Bhat

B.Tech. in Civil Engineering

July 2012 – June 2016

National Institute of Technology Srinagar, India

Research Experience

Research Associate

June 2025 – Present

Indian Institute of Technology Roorkee, India

Project: *Enhancing Sustainability of RC Frame Buildings through Seismic Safety and Design Life Elongation Using GFRP Reinforcement*

Funded Projects

Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels

Funding: INR 38 Lakh, NBCC (India) Limited

Institution: Indian Institute of Technology Roorkee, India

Publications

- Bhat, Z. M., & Singh, Y. (2025). Seismic risk assessment of fly ash brick and AAC block masonry infilled RC frame buildings designed for modern codes. *Journal of Earthquake Engineering*, 29(12), 2577–2601. <https://doi.org/10.1080/13632469.2025.2523852>
- Bhat, Z. M., & Singh, Y. (2025). Out-of-plane behavior of masonry infills of different types and slenderness ratios under reversed cyclic loading. *Journal of Structural Engineering*, 151(6), 04025056. <https://doi.org/10.1061/JSENDH.STENG-13761>
- Bhat, Z. M., Singh, Y., & Agarwal, P. (2024). Cyclic testing and diagonal strut modelling of different types of masonry infills in reinforced concrete frames designed for modern codes. *Engineering Structures*, 317, 118695. <https://doi.org/10.1016/j.engstruct.2024.118695>
- Bhat, Z. M., Singh, Y., & Agarwal, P. (2023). Characterization of mechanical behavior of different types of masonry with a detailed investigation of full-field strain using digital image correlation. *Construction and Building Materials*, 407, 133502. <https://doi.org/10.1016/j.conbuildmat.2023.133502>

- Waseem, S. A., **Bhat, Z. M.**, & Bhat, J. A. (2022). An experimental investigation into the behavior of steel-timber composite beams. *Practice Periodical on Structural Design and Construction*, 27(1), 04021055. [https://doi.org/10.1061/\(ASCE\)SC.1943-5576.0000636](https://doi.org/10.1061/(ASCE)SC.1943-5576.0000636)

Under Review

- **Bhat, Z. M.**, & Singh, Y. Seismic fragility analysis of masonry infills in RC buildings in the out-of-plane direction considering in-plane damage. *Journal of Building Engineering*. (First round of review completed).
- **Bhat, Z. M.**, & Singh, Y. Cyclic behavior and damage assessment of masonry infilled RC frames with openings using digital image correlation. *Engineering Structures*.

International Conferences

- **Bhat, Z. M.**, & Singh, Y. (2025). Reversed out-of-plane cyclic testing of masonry infills using double-leaf airbags. In *COMPDYN 2025*, Rhodes Island, Greece, 15–18 June 2025. <https://doi.org/10.7712/120125.12458.24591>
- **Bhat, Z. M.**, & Singh, Y. (2025). Cyclic in-plane behavior of masonry infilled RC frames with detailed interpretation of damage using digital image correlation. In *15th Canadian Masonry Symposium*, Ottawa, Canada, 2–5 June 2025.

Technical Reports

- **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2024). *Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels*. NBCC (India) Limited, India.
- **Team Member**, *Compendium of Traditional Earthquake-Resilient Construction Practices*, NDMA, Government of India, 2022. [Report Link](#)

Data and Software Contributions

- **Bhat, Z. M.**, & Singh, Y. (2024). Fly ash brick and AAC block masonry infilled RC frame buildings: OpenSees models and responses. *DesignSafe-CI*. <https://doi.org/10.17603/ds2-4acg-tt80>

Technical Skills

- **Programming & Tools:** Python, MATLAB, Tcl, LaTeX
- **Software:** OpenSees, STKO, ETABS, SAP2000, OpenQuake
- **Experimental Techniques:** Digital Image Correlation (DIC), Laboratory Instrumentation and Data Acquisition, Servo-hydraulic Actuator Control, Load Cell Calibration

Awards and Achievements

- MHRD Fellowship for doctoral studies – Government of India
- MHRD Fellowship for M.Tech studies – Government of India
- Ranked in top 2% in GATE 2016 among 119,873 candidates

Mentorship Experience

- **Adil Ahmed** (Ph.D. Student) – *Seismic Safety of EPS Core RC Sandwich Panel Buildings*; Mentored in experimental testing, test setup design, and numerical modeling.
- **Ravi Shastri** (Master's Student) – *Risk-Targeted Seismic Hazard Assessment in the Indian Himalayas*; Guided in OpenQuake-based hazard analysis and seismic risk assessment.

Workshops and Courses

- **6th International Course on Seismic Analysis of Structures** 19–22 July 2021
University of Palermo, Italy
- **Nonlinear Modelling of RC Structures Using OpenSees** 22–26 June 2020
Università degli Studi di Napoli Federico II, Italy

Teaching Assistantships

- **Advanced Earthquake Resistant Design of Structures (EQN-512)** IIT Roorkee
Instructor: Prof. Yogendra Singh
- **Earthquake Resistant Design of Structures (EQN-563)** IIT Roorkee
Instructor: Prof. Yogendra Singh

References

Prof. Yogendra Singh

Professor, Department of Earthquake Engineering, IIT Roorkee, India

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Prof. Pankaj Agarwal

Professor, Department of Earthquake Engineering, IIT Roorkee, India

Email: pankaj.agrawal@eq.iitr.ac.in; **Phone:** +91-9837004618; **Website:** [Google Scholar](#)

Prof. Pradeep Bhargava

Professor, Department of Civil Engineering, IIT Roorkee, India

Email: p.bhargava@ce.iitr.ac.in; **Phone:** +91-9837204501; **Website:** [Google Scholar](#)